# Rural Credit Risk Analysis & Decision Support System

GROUP C-5 | NEWTON SCHOOL OF TECHNOLOGY

A data-driven framework to assess creditworthiness for rural borrowers without traditional credit scores, built using Python, Tableau Public, and GitHub.

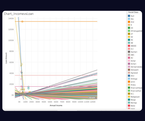
**Sector**
Rural Finance & Banking

**Stakeholders**
Regional Bank Managers & Credit Officers

**Dataset**
40,000 raw records processed from Kaggle

**Tech Stack**
Python (Pandas, SciPy), Tableau Public, GitHub

## The Business Challenge



### Credit Gap in Rural India

Rural populations across India face a systemic **"credit gap"** — the absence of formal CIBIL scores leaves millions of creditworthy borrowers invisible to traditional banking systems.

Bank managers currently rely heavily on intuition, leading to two costly outcomes:

- **Unfair loan rejections** for viable borrowers who lack formal credit history
- **High default rates (NPAs)** when loans are approved without rigorous risk assessment

Our goal: Identify non-traditional socio-economic proxies to accurately segment borrowers into risk categories.

## Data Engineering & ETL Pipeline

- Data Sourcing
- Data Cleaning
- Optimisation
- Ready for Visualisation

The pipeline standardised inconsistent categorical labels (e.g., 'Mouchi' → 'Mochi'), handled missing values via Median Imputation, and capped annual income outliers at the 95th percentile to prevent skew — ensuring high-performance interactive visualisation.

## Business KPI Framework

**Disposable Annual Income**
Annual Income − (Monthly Expenses × 12). Represents actual cash available for loan repayment.

**Debt-to-Income (DTI) Ratio**
(Loan Amount ÷ Annual Income) × 100. Measures the financial burden on the borrower.

### Risk Segmentation by DTI

- 🟢 **Low Risk** — DTI < 15%
- 🟡 **Medium Risk** — DTI 15–30%
- 🔴 **High Risk** — DTI > 30%

## Exploratory Data Analysis (EDA)

### Sectoral Analysis

Identified which primary businesses (e.g., Tailoring, Agriculture) receive the most funding, revealing concentration risk in specific sectors.

### Geographic Distribution

Mapped loan volume across different rural hubs to identify regional concentration risk and high-density default zones.

### Income Trends

Analysed the distribution of annual income to understand the average borrower profile and identify outlier segments.

**Key Finding:** A strong concentration of loans in specific sectors increases systemic risk if that sector faces a downturn.

## Advanced Statistical Analysis

### Statistical Methods Applied

**Correlation Heatmap**
Found a strong positive correlation between Disposable Income and loan repayment capacity.

**Hypothesis Testing (T-Tests)**
Determined whether gender significantly impacts loan amounts, ensuring fairness and equity in lending decisions.

**Risk Distribution Analysis**
Quantified the proportion of the current portfolio falling into the "High Risk" DTI category.

**Key Insight:** Disposable income is a far more accurate predictor of repayment than gross annual income.

## Interactive Tableau Solution

### Dashboard Architecture

**Executive View**
Top-level KPI cards: Total Borrowers, Average Income, Total Loan Volume

**Operational View**
Geographic map for regional drill-downs paired with sectoral bar charts

**Global Filters**
Social Class, City filters and Map-as-Filter for real-time risk auditing

## High-Value Business Insights

**1 — The Risk Gap**
Borrowers with **negative disposable income** are the most likely to default, regardless of their social class — a critical blind spot in traditional assessment.

**2 — Anomaly Detection**
Identified **"High-Risk" outliers** — low-income borrowers receiving disproportionately high-value loans — flagged for immediate portfolio review.

**3 — Sector Volatility**
Certain rural businesses show **higher volatility** in repayment capacity compared to others, requiring sector-specific risk weighting.

**4 — Geographic Risk**
Specific districts show a **higher density of "High Risk" segments**, enabling targeted intervention and monitoring by regional officers.

## Strategic Recommendations

### Implementation Rationale

- **Risk-Based Pricing:** A tiered interest rate system offsets potential losses from High-Risk segments whilst remaining competitive for Low-Risk borrowers.
- **Collateral Requirements:** Mandatory co-signers or physical assets for borrowers with negative disposable income provide a safety net against default.
- **Portfolio Diversification:** Capping loan exposure in over-saturated sectors and incentivising emerging rural micro-entrepreneurs reduces systemic concentration risk.

## Expected Impact & Conclusion

**10–15%**
**NPA Reduction**
Projected reduction in Non-Performing Assets through evidence-based risk segmentation

**10K**
**Records Analysed**
Optimised dataset enabling high-performance, real-time interactive visualisation

**3**
**Risk Tiers**
Clear Low, Medium, and High risk categories replacing intuition-based decisions

> Moving from **intuition-based lending** to **evidence-based intelligence** — increasing financial inclusion for creditworthy rural borrowers whilst protecting the bank's portfolio from avoidable defaults.

**💰 Financial Impact**
10–15% projected reduction in NPAs through DTI-driven risk assessment

**⚡ Operational Efficiency**
Faster, automated loan approvals based on the DTI and Risk Level framework

**🤝 Social Value**
Increased financial inclusion for "Low Risk" borrowers previously ignored by traditional banks

Made with Gamma